Crypt-Arithmetic Problem

AIM:

To write a Python program to solve a Crypt-Arithmetic Problem by assigning unique digits to letters to satisfy a mathematical equation.

Algorithm:

1. Start the program.
2. Read the given arithmetic expression containing letters.
3. Identify all unique letters in the expression.
4. Assign different digits to the letters.
5. Ensure that leading letters of multi-digit numbers are not assigned zero.
6. Substitute the digits into the expression.
7. Check whether the arithmetic equation is valid.
8. Display the valid digit assignment if a solution exists.
9. Stop the program.

Code :

```python
import itertools

def solve_cryptarithmetic():
    letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
    digits = range(10)

    for perm in itertools.permutations(digits, len(letters)):
        mapping = dict(zip(letters, perm))

        if mapping['S'] == 0 or mapping['M'] == 0:
            continue

        send = mapping['S'] * 1000 + mapping['E'] * 100 + mapping['N'] * 10 + mapping['D']
        more = mapping['M'] * 1000 + mapping['O'] * 100 + mapping['R'] * 10 + mapping['E']
        money = (mapping['M'] * 10000 + mapping['O'] * 1000 +
                 mapping['N'] * 100 + mapping['E'] * 10 + mapping['Y'])

        if send + more == money:
            return mapping, send, more, money

    return None, None, None, None

mapping, send, more, money = solve_cryptarithmetic()

if mapping:
    print("SEND + MORE = MONEY Solution:")
    for letter in sorted(mapping.keys()):
        print(f"{letter} = {mapping[letter]}")
    print()
    print(f"  {send}")
    print(f"+ {more}")
    print("------")
    print(f" {money}")
else:
    print("No solution found.")
```

Output :

```
PS D:\CSA1702-AI\Experiments> & C:\python311\python.exe d:/CSA1702-AI/Experiments/exp04_cryptarithmetic.py
SEND + MORE = MONEY Solution:
D = 7
E = 5
M = 1
N = 6
O = 0
R = 8
S = 9
Y = 2

  9567
+ 1085
------
 10652
PS D:\CSA1702-AI\Experiments>
```

Result :

Thus, the Python program to solve the Crypt-Arithmetic Problem was executed successfully.